

Real Madrid CF

Tactical Dashboard

Transforming event-level football data into actionable tactical intelligence

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Data: Opta Stats Perform (event-level) · Python · Plotly · Streamlit

Live charts rendered from real match data · June 2026

The Problem & The Objective

- Elite football generates ~1,500–2,000 on-ball events per match — far too granular to read by eye.
- Traditional analysis suffers from outcome bias: a scoreline hides whether a performance was earned or lucky.
- Coaching staff need the data-to-decision layer that turns a raw Opta feed into tactical answers.
- Objective: quantify HOW Real Madrid play — not just whether they win — across attacking, defending, transitions and set pieces.
- Deliver an interactive, transparent, reproducible tool for match prep, opposition scouting and review.

KEY POINT

Every metric is traceable to an explicit formula over observed events — no synthetic data anywhere.

Dashboard Architecture — Seven Modules

- Overview — season KPIs, results, xG / xGA and form trends.
- Match Analysis — shot maps, xG curve, pass & build-up networks, transitions, set pieces.
- Player Analysis — squad table, per-player stats, multi-radar, touch heatmaps.
- Tactical Phases — PPDA, pressing maps, four-phase A/B/C/D scoring, recoveries, field tilt.
- Opponent Scout — opposition profiling, threat heatmap, set-piece tendencies, auto report.
- Benchmarking — cross-competition and cross-season comparison.
- Report — one-click PDF / DOCX export.

KEY POINT

All seven modules share one data pipeline and one set of KPI definitions — the same xG everywhere.

Data Foundation & Methodology

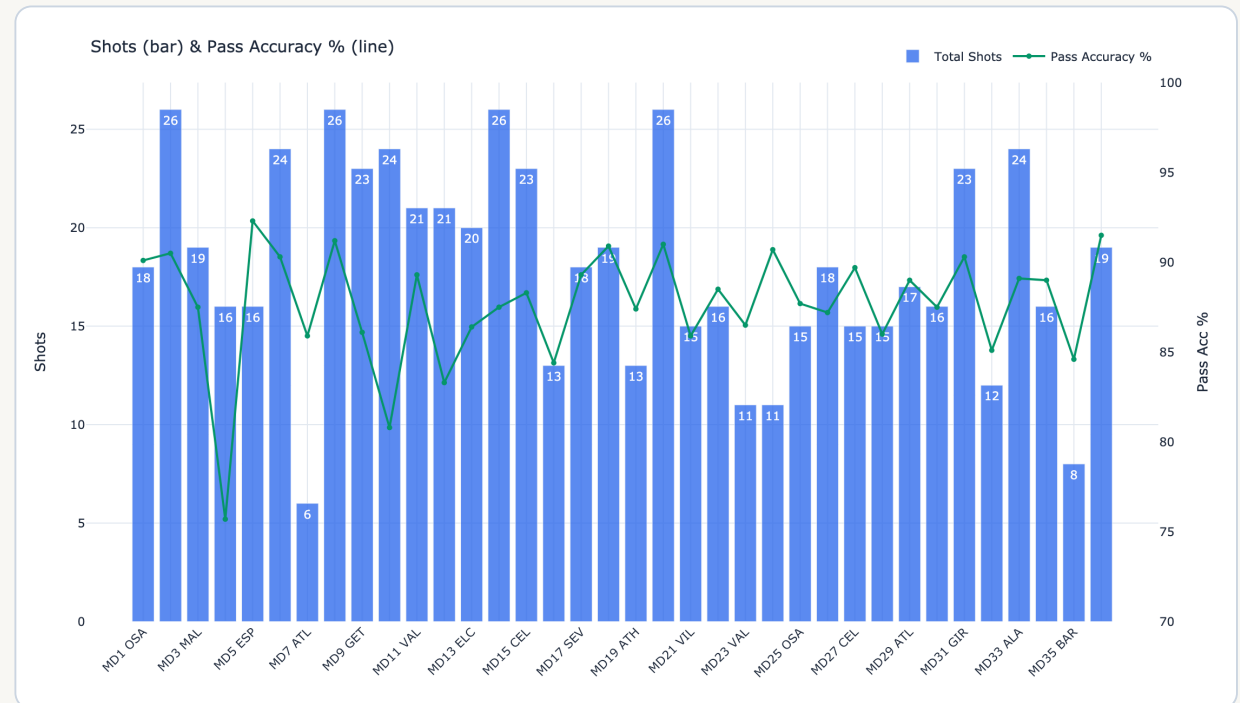
- Source: Opta Stats Perform event-level JSON — one file per match.
- Scope: 108 clean match files · 3 competitions (LaLiga · UCL · Copa del Rey) · 2 seasons (2024-25, 2025-26).
- Pipeline: parse & normalise events → compute match KPIs → aggregate to season / re-slice to player, phase, opponent.
- Transparent models: positional xG (distance + angle), PPDA (pressing intensity), field tilt, four-phase scoring.
- Possession is a documented pass-share proxy; no tracking data — limitations stated openly.

KEY POINT

Showcase fixture for the visuals that follow: Real Madrid 2–1 Barcelona, LaLiga 2025–26, Matchday 10.

Overview — Season Context

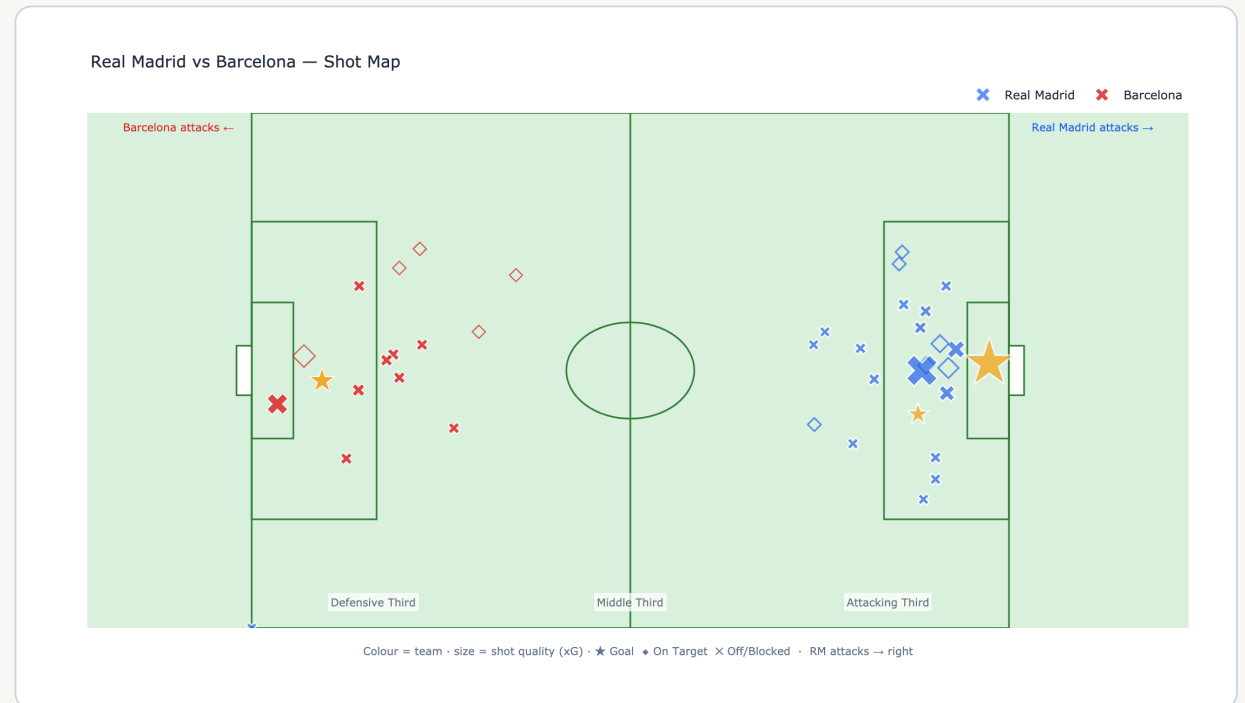
- Tracks process metrics (shot volume, pass accuracy) across the season, not just results.
- Process metrics are more stable and more predictive than outcomes.
- Falling shots with high pass accuracy = early warning of sterile possession.
- The panel a coach watches weekly to spot a problem before the table shows it.



Overview module — Shots & Pass-Accuracy Trend, rendered live from the real LaLiga 2025-26 feed.

Match Analysis — Shot Map

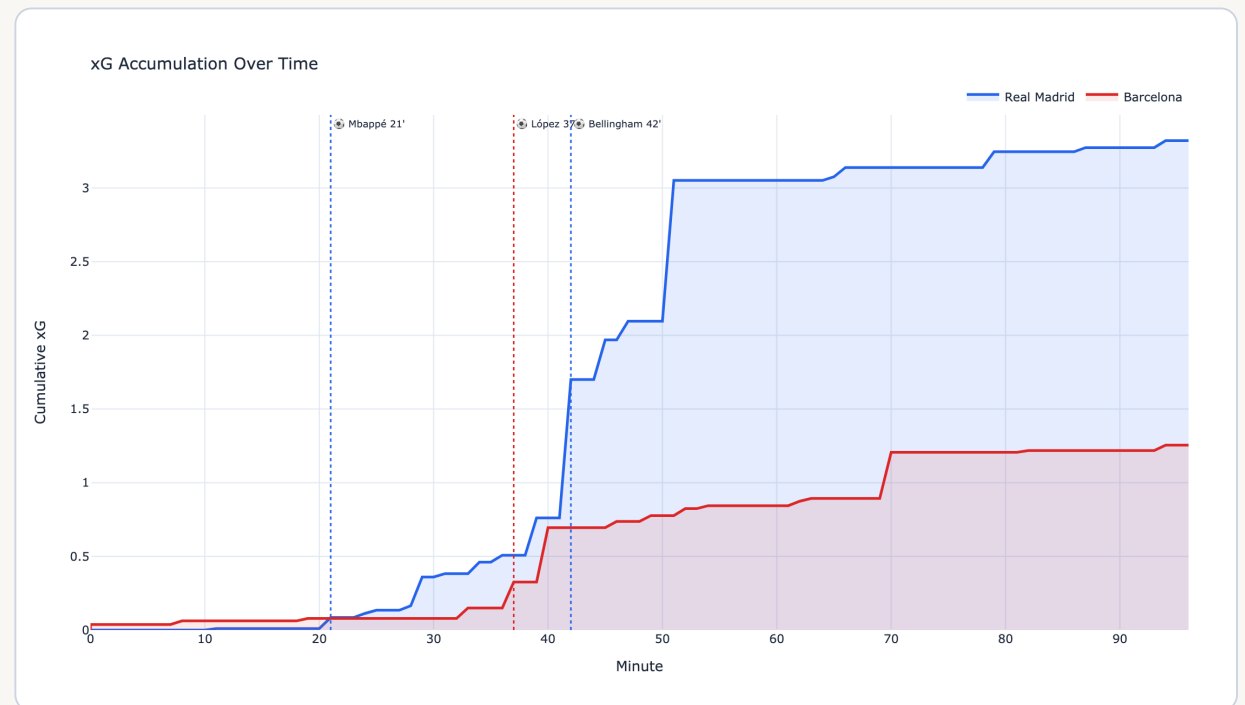
- Every shot at its true location; marker size $\propto$ xG; ★ = goal.
- Real Madrid (blue) attack right; Barcelona (red) attack left.
- RM's chances cluster centrally inside the box — high volume AND high quality.
- Answers: where, and how good, were the chances — for both teams.



Match Analysis — live Shot Map for Real Madrid 2-1 Barcelona (24 shots · 2 goals · 3.34 xG).

Match Analysis — xG Accumulation

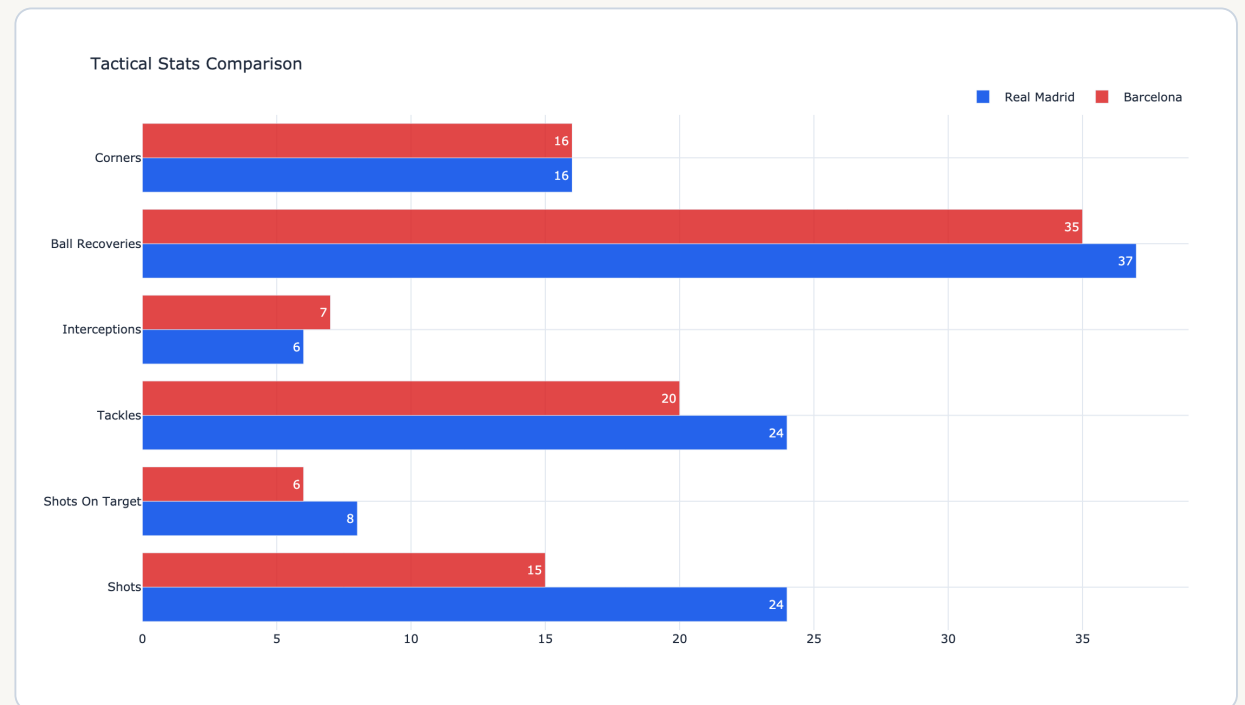
- Running total of Expected Goals, minute by minute, for both teams.
- Slope = rate of chance creation; a tall step = a single big chance.
- Reveals WHEN control happened — early dominance vs a nervy finish.
- Full-time gap between the lines = the match's xG difference.



Match Analysis — Expected Goals accumulating through the match from the real event feed.

Match Analysis — Tactical Comparison

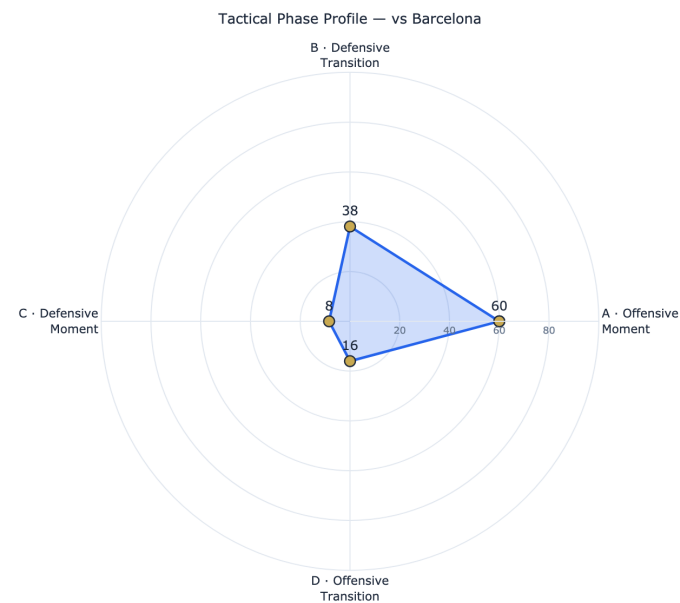
- Head-to-head bars of the headline KPIs: possession, shots, passing, defensive actions.
- Built for communication speed — 'who won which battle' in one glance.
- Each metric read in RELATIVE context (the gap to the opponent, not the raw value).



Match Analysis — Tactical Comparison, Real Madrid vs Barcelona, from the real feed.

Tactical Phase Profile (A/B/C/D)

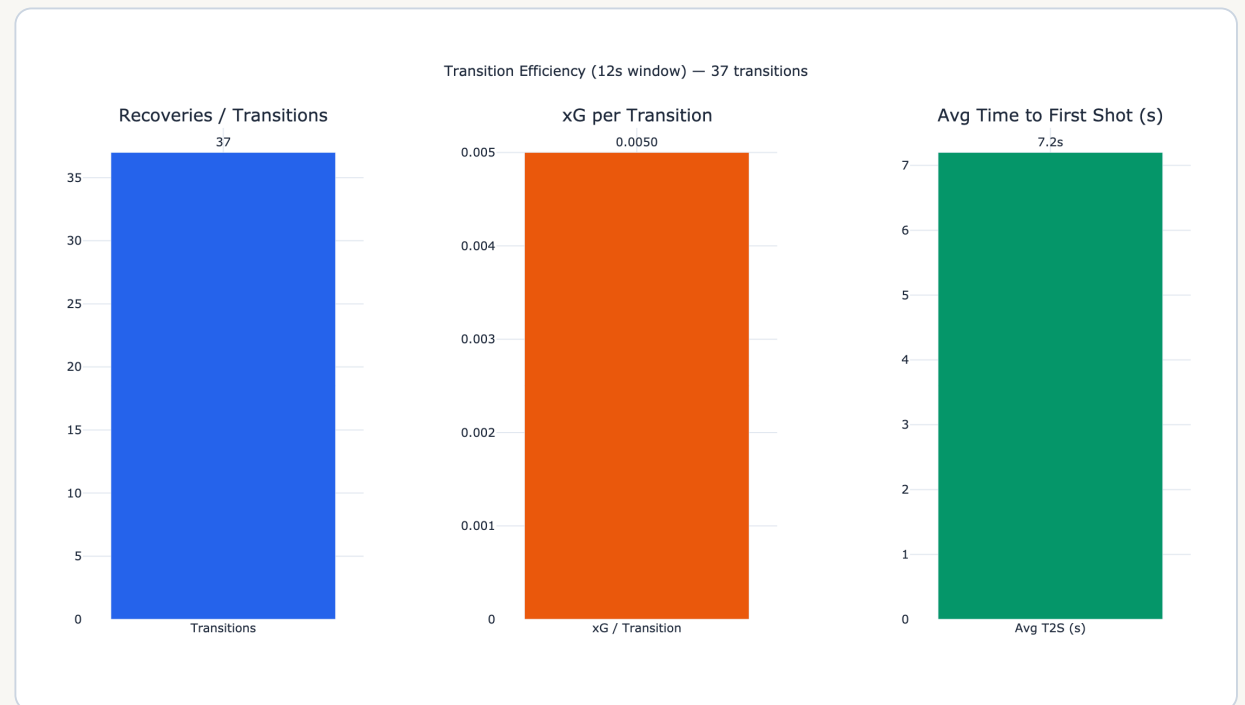
- Four moments of play scored 0-100: Offensive Moment (A), Defensive Transition (B), Defensive Moment (C), Offensive Transition (D).
- The radar 'shape' is the match's tactical fingerprint.
- This fixture: strong A + B, lower C + D = a possession-and-counter-press game.
- Composite indices, grounded in real PPDA, xG, recoveries & transition rate.



Tactical Phase Profile radar — the dashboard's executive summary of how the team played.

Attacking Transitions

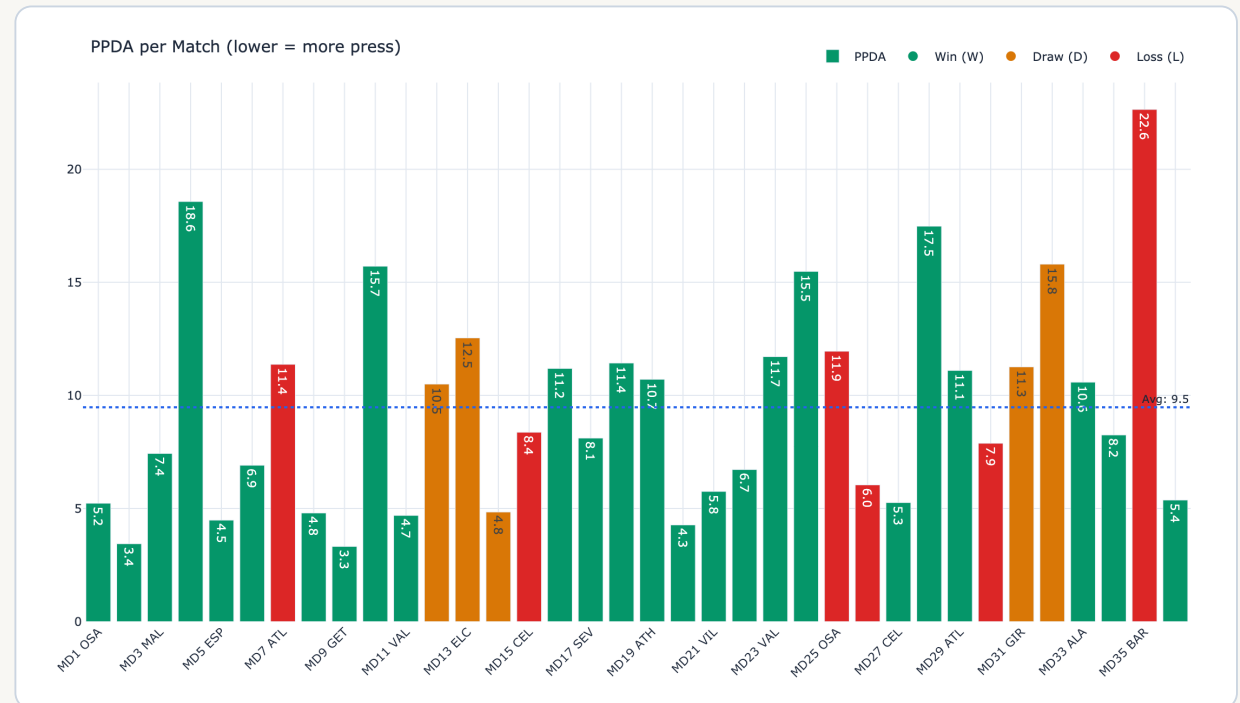
- Locates every ball regain, then tests for a shot within the time window (5-15 s).
- Quantifies fast-break efficiency and transition xG — the counter-attack as a number.
- Against possession sides, transitions punish their committed players.
- Window is an analyst control, not a hidden assumption.



Match Analysis — Transition Metrics, quantifying Real Madrid's counter-attacking threat.

Pressing Intensity — PPDA Trend

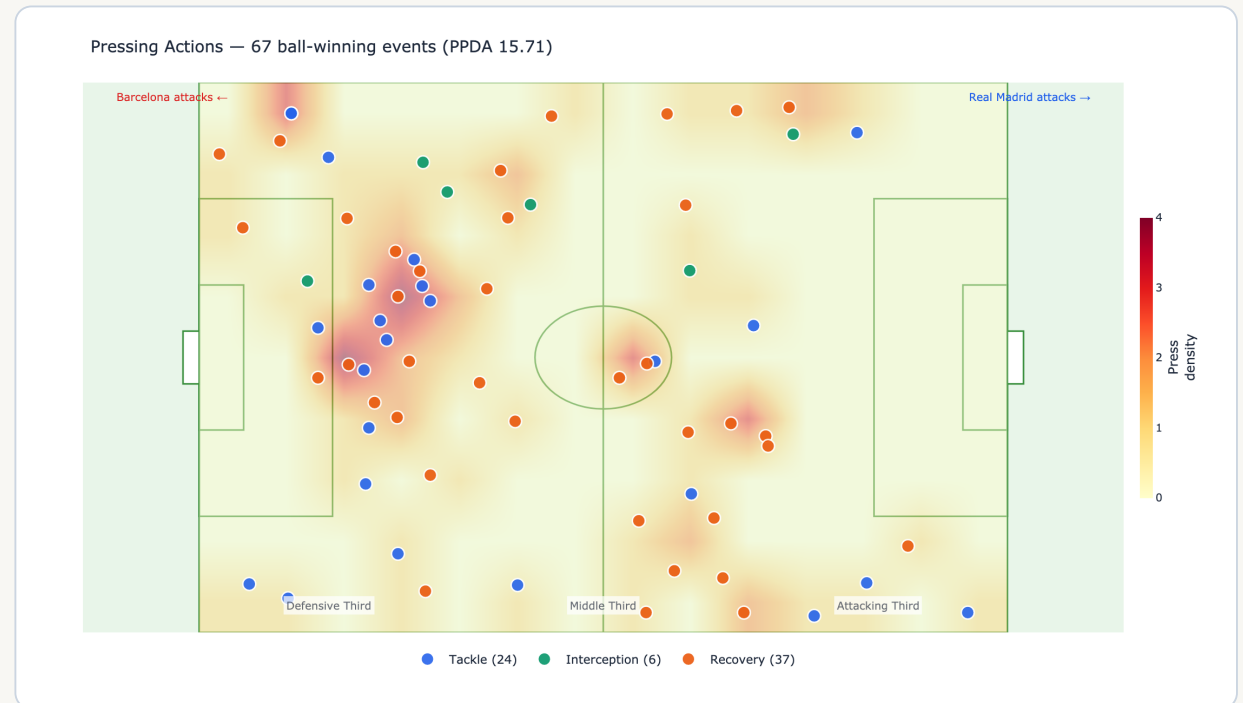
- PPDA = passes the opponent completes per RM defensive action (lower = more intense press).
- Plotted every matchday with the season average marked.
- Tight band = a drilled pressing identity; wide scatter = game-state dependent.
- The instant verdict on whether 'we pressed well today' is actually true.



Tactical Phases — season-long PPDA trend across the real LaLiga campaign.

Pressing Actions Map

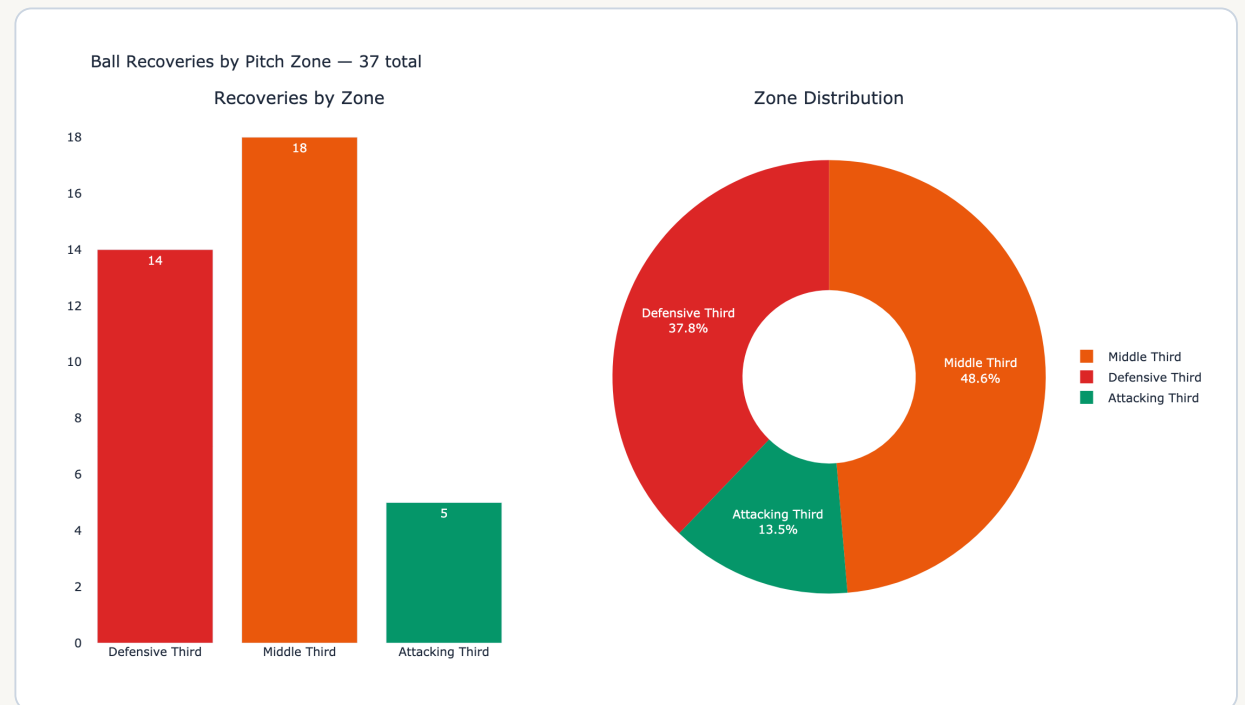
- Where on the pitch Real Madrid engaged defensively — every press, tackle, interception.
- Reveals pressing HEIGHT: high press vs mid-block vs low block.
- Exposes asymmetry — a flank pressed harder is a flank the opponent will exploit.
- Direct evidence of whether the pressing instruction was executed.



Tactical Phases — Pressing Actions Map for the real match.

Ball Recoveries by Zone

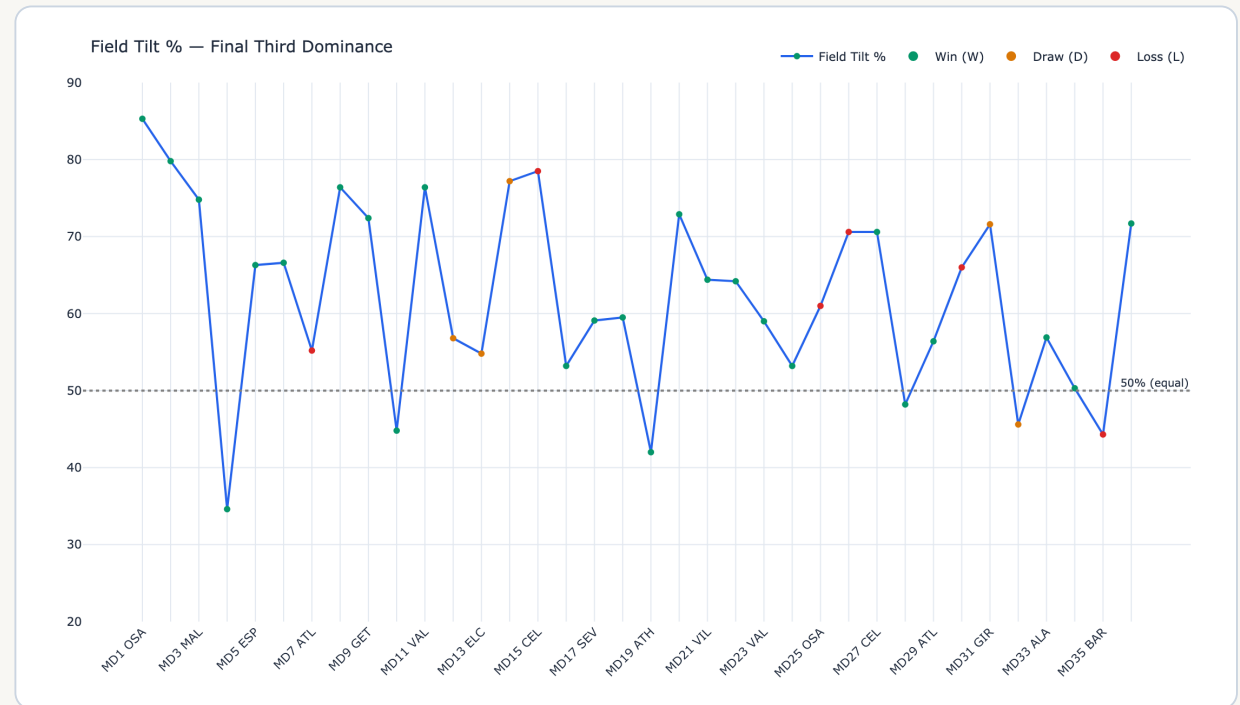
- WHERE the team actually wins the ball back — defensive / middle / attacking third.
- This match: 37 recoveries — Mid 48.6% · Def 37.8% · Att 13.5% (a mid-block profile).
- Attacking-third recoveries are the most dangerous regains in football.
- Read against intention: great for a mid-block, disappointing for a high press.



Tactical Phases — Ball Recoveries by Pitch Zone (bar + distribution) from the real feed.

Territorial Dominance — Field Tilt

- Share of all final-third activity belonging to Real Madrid.
- The antidote to misreading possession — control that becomes territory.
- 55% possession with 75% field tilt beats 60% possession with 51% tilt.
- Validates the PURPOSE of keeping the ball.

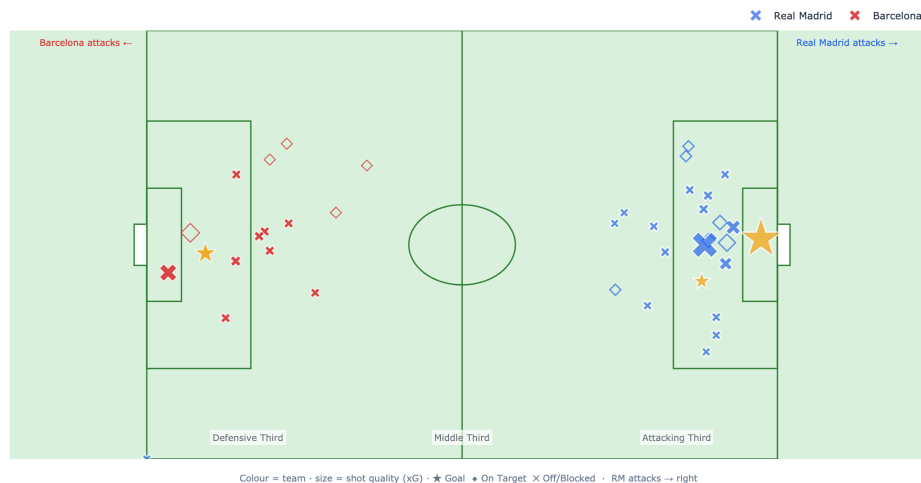


Tactical Phases — Field Tilt across the real season, a measure of territorial dominance.

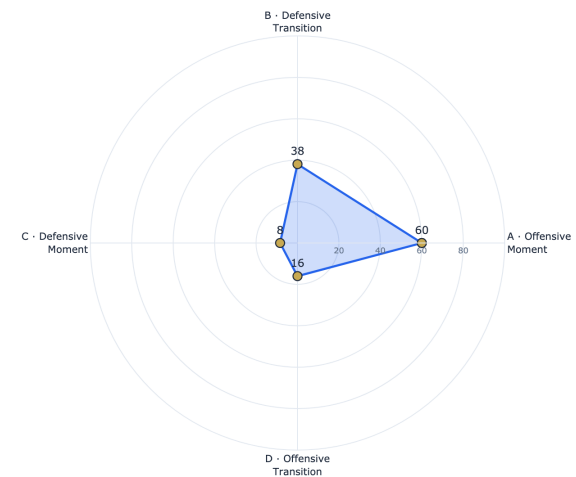
How the Panels Triangulate

- No single panel proves a conclusion — together they make it unarguable: the Shot Map shows high-quality central chances, the radar collapses the match into one possession-and-counter-press shape.
- The same triangulation runs across xG, field tilt, recoveries and PPDA — a tactical claim that is true leaves its fingerprint on several independent panels at once.

Real Madrid vs Barcelona — Shot Map



Tactical Phase Profile — vs Barcelona



Two independent views of the same real match converging on one tactical read.

Implementation Status

- All nine core analysis features are IMPLEMENTED, each with dedicated panels:
 - Pre-Match · Attacking Transitions · Organized Attack · Defensive Strategy · Set Pieces.
 - Possession Recovery · Post-Match · Final-Third Entries · Match Info.
- Plus extras: Zone 14, Crossing, Progressive Passes, GK Distribution, Goalmouth, Penalty Analysis, Team & Player radars, Opponent Threat Heatmap, full Benchmarking suite, PDF/DOCX export.
- Deployed publicly with GitHub → Render CI/CD.

KEY POINT

Partial / Planned applies only to enhancements (shots-conceded heatmap, block-height viz, xT) — not to built features.

Limitations & Future Scope

- Event data only — no tracking: off-ball shape & pressing distances are approximated, not measured.
- xG is a transparent positional model (location only), not a provider post-shot model.
- Phase scores are interpretive 0–100 composites, not physical units.
- Descriptive & diagnostic — not predictive, not real-time.
- Roadmap: tracking-data integration · player per-90 depth · Expected Threat (xT) · ML pattern clustering · predictive analytics · video integration · live in-match dashboards.

KEY POINT

The platform is a foundation for continued research and professional deployment — not a finished endpoint.

From Data to Tactical Decision

One transparent, interactive platform — match prep, scouting, review and player development.

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